

PENETRATION TESTING REPORT

Footprinting & Network Scanning Phases

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program / Batch	B083 – Networkwalks
Course	Cybersecurity & Ethical Hacking
Modules Completed	W2-PM4 (The Harvester) W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks (written permission secured) 2. Own local LAN / lab network
Permission Secured?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phases 3–5: In Progress

1. Liability Disclaimer

This penetration testing activity was performed exclusively on systems and networks for which explicit written authorization had been obtained, or on the author's own local laboratory network. The scanning and reconnaissance exercises described in this report were conducted solely for educational purposes as part of the Networkwalks Cybersecurity Internship.

No production systems belonging to third parties were targeted without prior consent. The author and Networkwalks accept no liability for any misuse of the techniques, tools, or findings presented herein. All activities remained within the authorized scope defined for this practical assignment.

2. Introduction

This report documents the Week 2 practical activities completed for the Networkwalks Cybersecurity Internship. The focus of the week was footprinting, open-source intelligence (OSINT) collection, and network scanning / host discovery.

Two primary modules were completed:

- W2-PM4 – Passive reconnaissance using The Harvester on an authorized target domain.
- W2-PM5 – Active network discovery using Zenmap (Nmap GUI) against the author's own local / VirtualBox laboratory network.

The work identified public-facing digital footprint information, local IP addressing and subnet details, live hosts, associated MAC addresses, and a visual network topology. All findings are presented strictly as educational observations; no exploitation or vulnerability validation was performed.

3. Tools Used

The following tools were employed during the footprinting and scanning phases:

Tool	Purpose
The Harvester	Collect publicly available information (emails, hostnames, subdomains) associated with an authorized domain.
Zenmap (Nmap GUI)	Perform Ping Scan on the local subnet to discover live hosts, IP addresses, and MAC addresses; generate topology visualization.

Windows CMD / ipconfig	Identify local IP address, subnet mask, and default gateway of the host machine.
Kali Linux utilities	Supporting OSINT tools (WhatWeb, Nslookup, Curl, Wafw00f, DNSRecon) used during the broader footprinting phase.

4. Activities Performed

4.1 OSINT Footprinting & Reconnaissance

Passive reconnaissance was performed with The Harvester inside Kali Linux to gather publicly indexed information about the authorized target domain. The goal was to understand the organization's external digital footprint without any attempt to exploit or gain unauthorized access.

After reviewing the tool's help options, The Harvester was executed against the target domain using a selected public data source. Results typically included email addresses, hostnames, subdomains, and related IP information, depending on what had been publicly indexed.

All collected data were reviewed and documented. Any sensitive or personally identifiable information discovered during the exercise was redacted before inclusion in this report or any shared evidence. The activity illustrated how an adversary can build an initial profile of an organization solely from publicly available sources, and how security professionals can use the same techniques to measure external information exposure.

Figure 1 — The Harvester help output (Kali Linux)

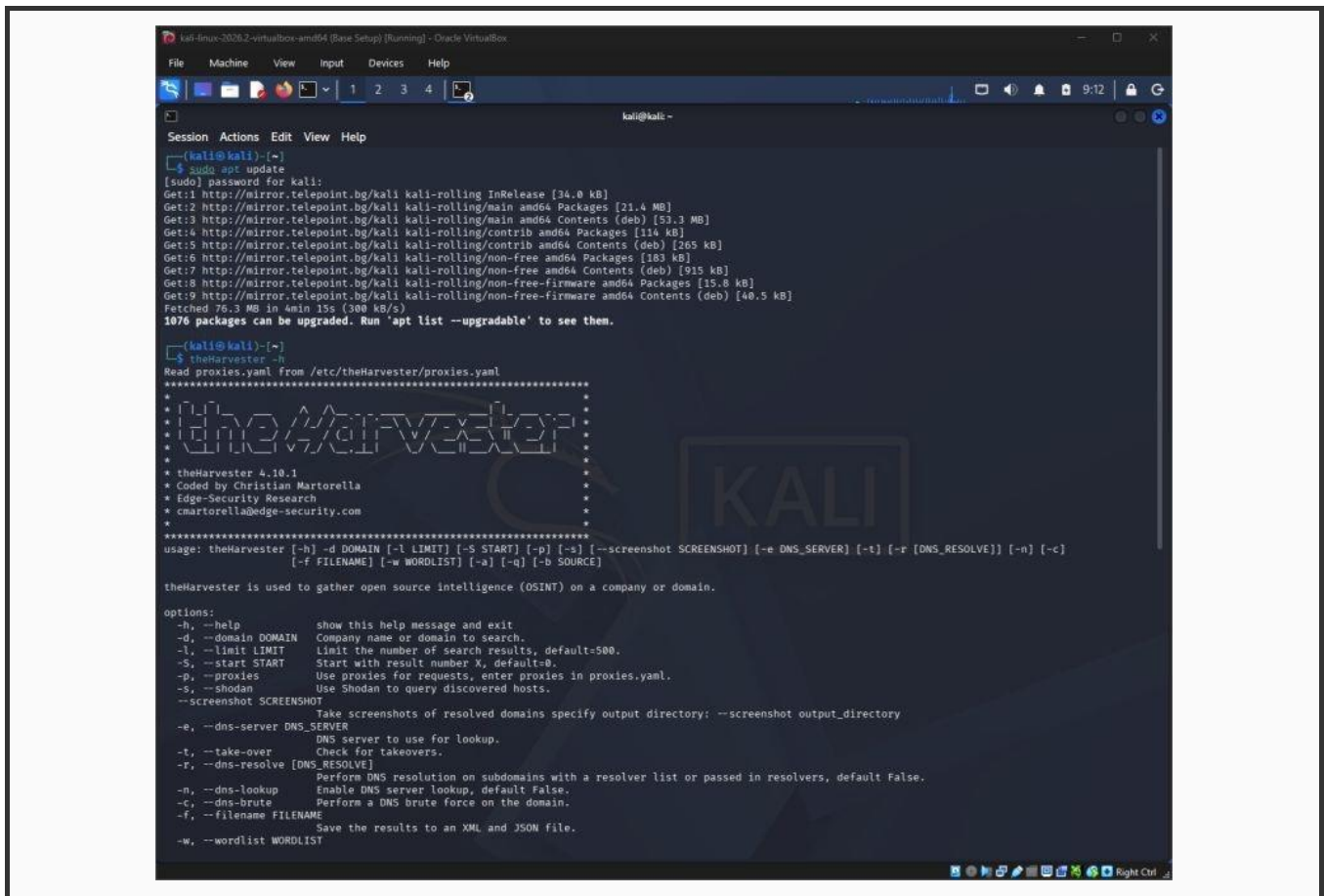


Figure 1: The Harvester tool help screen showing available OSINT options

4.2 Network Scanning with Zenmap

For the network-discovery module, Zenmap was used on the author's own local laboratory network. The practical required identification of the local IP address and subnet, discovery of live hosts, collection of IP and MAC address information, and generation of a network topology map.

Windows ipconfig was first used to determine the host's local IP address and LAN subnet. That subnet was then entered into Zenmap and a Ping Scan was executed to locate active hosts. After the scan completed, the Topology view was opened, the legend enabled, and the resulting diagram exported as required.

Note: Actual subnet ranges, host counts, and addresses in the final submission reflect the author's own laboratory environment rather than any sample values.

Figure 2 — Zenmap Ping Scan results

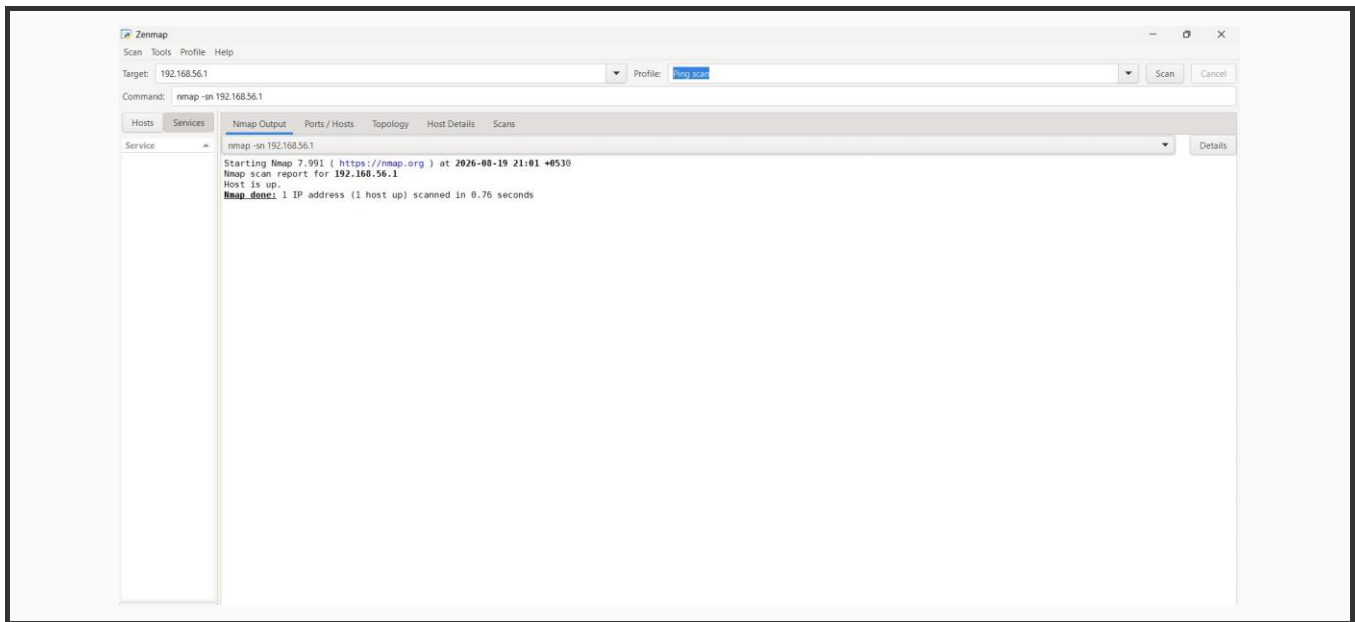


Figure 2: Zenmap Nmap Output showing Ping Scan of 192.168.56.1

Figure 3 — Zenmap Network Topology

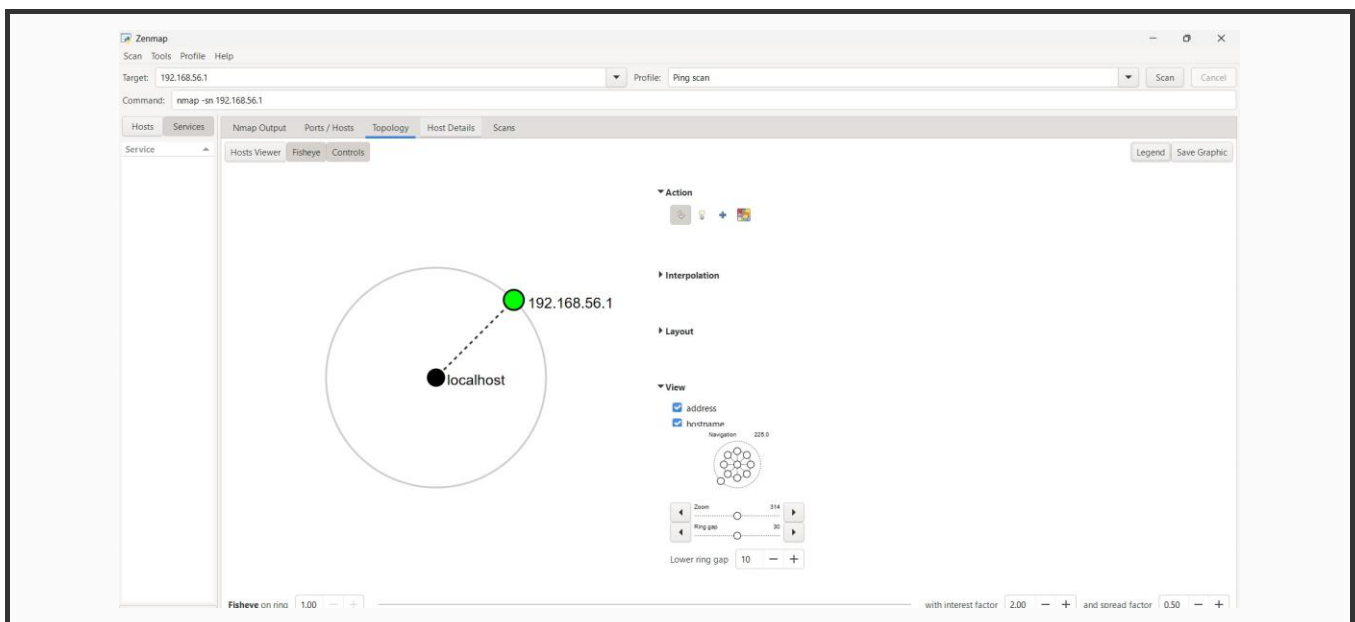


Figure 3: Zenmap Topology view showing localhost and discovered host 192.168.56.1

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network-scanning activities, the following potential risks were identified. These are observations from information-gathering exercises only; they do not constitute confirmed vulnerabilities.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
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1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version data to target known weaknesses	Medium
2	Server IP address identifiable	Nslookup resolved the domain to a public IP	Reveals network location of the web service	Low
3	HTTP technical information exposed	Curl returned response headers and exposed /wp-json/	Assists technology fingerprinting and further enumeration	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals details of the web application's security architecture	Low
5	DNS infrastructure information exposed	DNSRecon returned DNS, mail and service-related records	Helps build a broader infrastructure profile	Medium

Risk level key: Critical · Medium · Low

The practical exercises involved information gathering and host discovery only. No exploitation or vulnerability validation was performed. Consequently, the mere presence of a software version, IP address, or DNS record does not by itself indicate that a system is vulnerable. Further authorized security testing would be required to confirm any actual weakness.

6. Recommendations

Based on the observations from these activities, the following security improvements are recommended:

1. Review publicly exposed technology information

Organizations should regularly assess what information about their web technologies, CMS platforms, and plugins is publicly visible.

2. Keep software updated

CMS platforms, plugins, and other web technologies should be kept current and reviewed against published security advisories.

3. Review HTTP response headers

Headers should be examined to ensure that unnecessary technical details are not being disclosed.

4. Review DNS records regularly

DNS records should be audited periodically so that only required services remain publicly exposed.

5. Properly configure and monitor the WAF

Keep the Web Application Firewall (e.g., ModSecurity) enabled and tuned; it already provides a useful first line of defense.

6. Perform regular internal network discovery

Organizations should periodically scan their own networks to identify active devices and unexpected hosts.

7. Investigate unknown devices

Any unexpected device discovered during network scanning should be investigated and verified.

8. Maintain accurate network documentation

Network topology and device inventories should be documented and kept up to date.

9. Perform security testing only with authorization

Reconnaissance and scanning must always be limited to systems and networks for which appropriate written authorization has been obtained.

7. Conclusion

During Week 2 of the Cybersecurity & Ethical Hacking internship, practical activities covering footprinting, reconnaissance, and network scanning were completed successfully.

In the footprinting phase, multiple Kali Linux tools were used to collect information about the authorized target domain. WHOIS supplied registration data, WhatWeb identified web technologies, Nslookup resolved domain names, Curl inspected HTTP headers, Wafw00f detected the presence of a WAF, and DNSRecon provided additional DNS infrastructure details.

In the network-scanning phase, Zenmap was used to identify the local network configuration, discover active hosts, collect IP and MAC address information, and produce a network topology diagram.

These exercises reinforced that thorough information gathering is a foundational skill in cybersecurity. Even before any exploitation is attempted, a security professional can learn a great deal about an environment by carefully analyzing publicly available data and network responses. Findings must be documented clearly—what was performed, what was observed, what the observation implies, the potential risk, and recommended mitigations.

Finally, all reconnaissance and scanning activities were conducted strictly within an authorized educational scope. The same discipline of scope limitation and written permission must be observed in any real-world engagement.

8. Evidences Collected

The following evidence screenshots are included in this report (each presented with a border for clarity):

- Figure 1 — The Harvester help / usage screen (Kali Linux)
- Figure 2 — Zenmap Ping Scan Nmap Output
- Figure 3 — Zenmap Network Topology visualization

Additional supporting files (exported topology PDF, raw scan logs) are retained as part of the practical submission package.

— *End of Report* —